

RELTWIN: CONTRASTING LOCALLY VALID TIMESTAMP ANSWERS FOR QUERY-SPECIFIC AUDIO GROUNDING

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ABSTRACT

Audio temporal grounding must distinguish which event occurrence answers a query, even when opposite event orders coexist in one recording. Returning the same window for inverse queries reveals a query–window binding error. We introduce RelTwin, an add-on fine-tuning method that constructs opposite-order windows from shared excerpts and contrasts their timestamp answers. Both windows have acoustic support; the query determines the correct answer. RelTwin combines answer competition, exchange consistency, and set-level preferences while preserving direct timestamp generation. On SpotSound-Bench, Clotho-Moment, and UnAV-100, adapted SpotSound-A reaches 59.43, 86.68, and 74.03 mIoU; adapted TimeAudio exceeds published mIoU by 8.16, 5.48, and 14.48 points. Matched candidate-supervision experiments show 31.25 points of joint-localization gain on synthetic relation-development data and 1.02 points of SpotSound mIoU gain. The binding advantage persists under timing intervention, supporting supervision of which occurrence answers the query.

Index Terms— Audio temporal grounding, timestamp answer contrasts, query–window binding, paired supervision

1. INTRODUCTION

Audio temporal grounding maps a query to event intervals [1, 2]. In a recording containing both “dog then rooster” and “rooster then dog,” the descriptions refer to different windows. Recognizing the categories establishes presence; answering the query requires selecting the requested order. Returning the same real window for both queries is a *query–window binding error*. We target this distinction between event detection and query-specific localization.

RelTwin (Fig. 1) addresses this binding problem through *locally valid timestamp contrasts*. Shared excerpts form opposite-order windows in one recording. Each window contains a real event sequence, but its correctness as an answer depends on the query. Sequence supervision rewards correct timestamp answers individually; the added candidate loss

explicitly compares the two answers under each query. Reversing the query exchanges their labels while keeping the recording fixed, directly training the model to choose the requested occurrence.

RelTwin fine-tunes an existing grounding model using its own timestamp-answer likelihoods. Query-exchange consistency aligns preferences under inverse queries; set-level preference optimization extends the comparison to coverage, event count, and boundaries. Inference remains direct timestamp generation. We evaluate grounding on three public benchmarks and isolate answer-level competition through matched training comparisons.

We make three contributions. First, we characterize query–window binding errors when inverse relations co-occur in one recording. Second, we introduce locally valid timestamp contrasts as an additional fine-tuning objective for query-specific answer selection. Third, public grounding evaluations, paired diagnosis, and a timing intervention connect the observed localization performance to improved query–window binding.

2. RELATED WORK

Audio temporal grounding. Weakly supervised grounding learns temporal correspondences from clip-level captions, reducing dependence on frame-level annotations [3]. Audio-language models such as Pengi [4], Qwen2-Audio [5], and Audio Flamingo 3 [6] provide audio-conditioned text generation. For temporal localization, TimeAudio introduces temporal markers and absolute time-aware encoding [7], while SpotSound interleaves timestamps with audio tokens and uses absent-event queries to suppress hallucinated intervals [8]. RelTwin targets query-specific selection when inverse relations coexist: both descriptions have acoustic support, and the requested relation determines the answer.

Compositional audio–language learning. CLAP aligns audio and text in a shared representation space [9]. T-CLAP introduces order-reversed descriptions and a temporal-focused contrastive objective [10]. CompA evaluates event order and attribute binding, and trains CompA-CLAP with composition-aware hard negatives and modular contrastive

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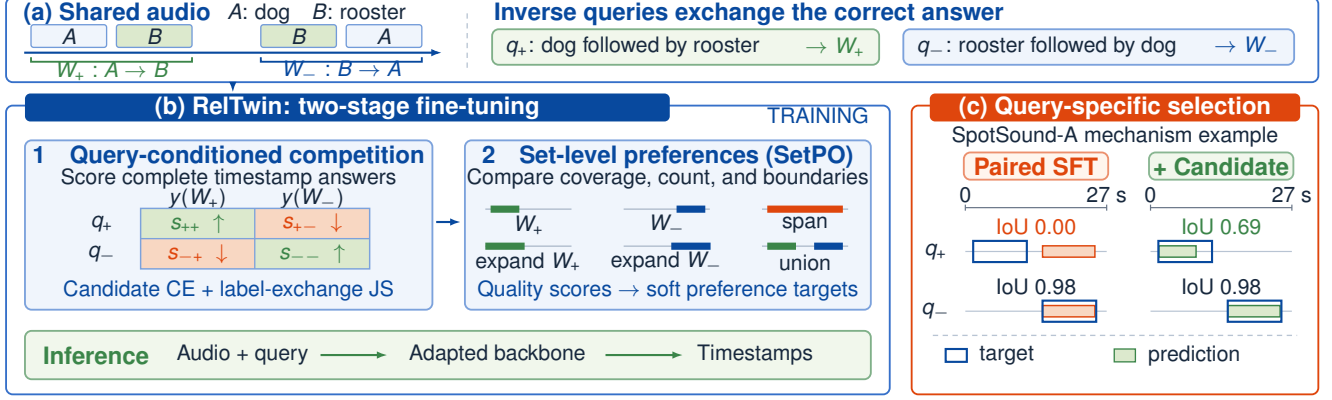


Fig. 1. RelTwin overview. (a) Shared excerpts yield opposite-order targets for inverse queries. (b) Candidate competition and exchange consistency precede SetPO; inference directly generates timestamps. Green/orange matrix cells mark matching/competing answers. (c) A SpotSound-A mechanism example shows candidate supervision recovering query-specific windows. Blue outlines denote targets; filled bars denote predictions, with set-IoU above.

learning [11]. CoSTALA combines spatio-temporal contrast with local event-caption alignment and feature consistency [12]. These methods learn compositional audio-text representations. RelTwin places supervision on complete timestamp answers: their query-conditioned likelihoods determine the preference between competing windows.

Hard negatives for temporal grounding. In compositional video grounding, SHINE combines generated hard-negative queries with within-video and cross-query saliency ranking [13]. RelTwin compares complete timestamp answers within one recording. Each denotes an event sequence with real acoustic support. Under a fixed query, one answer matches the requested order and the other answers the inverse relation; reversing the query exchanges their labels while preserving the audio context.

3. QUERY-SPECIFIC WINDOW LEARNING

3.1. Two true relations, two different answers

In one recording x , RelTwin arranges the same ESC-50 excerpts [14], A and B , as $A \rightarrow B$ and $B \rightarrow A$. Their local targets W_+ and W_- run from the first excerpt’s onset to the second’s offset, including the gap. Queries q_+ and q_- request the respective orders. Both windows have acoustic support, but W_- is positive for q_- and negative for q_+ ; reversing the query exchanges their labels.

The templates “followed by” and “and then” yield two inverse-query pairs per recording. Earlier-order layouts alternate (Section 5.3). Category rehearsal targets the union of both occurrences of A or B , retaining category-grounding supervision.

3.2. Contrast timestamp answers during training

We serialize an interval set S in the backbone’s timestamp format: SpotSound-A uses comma-separated spans such as from 1.500s to 11.750s; TimeAudio uses its native temporal tokens. The supervised answer tokens $y(S)$ exclude the prompt and are scored by

$$s_\theta(x, q, S) = \frac{1}{|y(S)|} \sum_t \log P_\theta(y_t | x, q, y_{<t}). \quad (1)$$

The 2×2 matrix M , with entries $M_{ij} = s_\theta(x, q_i, W_j)$ for $i, j \in \{+, -\}$, indexes queries by rows and answers by columns. Its diagonal labels $Y = I_2$ identify the correct query-answer pairs (Fig. 1b). Sequence supervision minimizes $\mathcal{L}_{\text{seq}} = -(M_{++} + M_{--})/2$, rewarding the two correct responses individually. It does not explicitly compare them with the other locally valid window. RelTwin trains the relative preferences $M_{++} > M_{+-}$ and $M_{--} > M_{-+}$ through row-wise probabilities $p_i(j) = \text{softmax}_j(M_{ij}/\tau)$, where $\tau > 0$. The candidate loss is

$$\mathcal{L}_{\text{cand}} = -\frac{1}{2}[\log p_+(+) + \log p_-(-)]. \quad (2)$$

Exchange Π swaps candidate labels; $\mathcal{L}_{\text{JS}} = \text{JS}(p_+, \Pi p_-)$ aligns their distributions. For rehearsal example (x_r, q_r, S_r) , define $\mathcal{R}_1 = -s_\theta(x_r, q_r, S_r)$. Stage 1 optimizes

$$\mathcal{L}_1 = \mathcal{L}_{\text{seq}} + \lambda(\mathcal{L}_{\text{cand}} + \gamma \mathcal{L}_{\text{JS}}) + \beta \mathcal{R}_1. \quad (3)$$

Set-level preference optimization (SetPO). Stage 2 compares six answers: $W_\pm$, their expanded versions, their enclosing span, and their union. Quality axes are set-IoU, precision-recall margins above IoU 0.3/0.5, count agreement, and duration-normalized boundary accuracy. Utility v_{ij} averages mean quality and net Pareto dominance (number dominated minus dominating, normalized by candidate count minus one). Targets $t_i = \text{softmax}(v_i/\tau_q)$ yield

Table 1. Audio temporal grounding (%; higher is better). Baselines: SpotSound Table 3 [8]. RelTwin adapts each backbone through the same two-stage framework; Δ is the percentage-point gain over its published baseline. Bold marks maxima among listed methods.

Method	SpotSound-Bench			Clotho-Moment			UnAV-100 subset		
	R1@.3	R1@.5	mIoU	R1@.3	R1@.5	mIoU	R1@.3	R1@.5	mIoU
<i>Non-LLM models</i>									
WTATG	53.0	34.5	38.4	12.1	6.3	9.1	53.0	37.0	38.4
AM-DETR	25.5	15.0	19.5	89.8	88.0	80.9	59.0	46.0	42.8
<i>Proprietary models</i>									
Gemini-2.5-Flash	34.8	27.8	25.7	45.7	33.7	36.9	48.0	37.0	35.6
Gemini-2.5-Pro	20.0	17.3	19.7	40.4	32.5	32.5	39.0	35.0	34.6
<i>Open-source models and RelTwin adaptation</i>									
Kimi-Audio	3.0	1.3	3.2	0.7	0.1	0.9	8.0	2.0	5.3
Qwen2-Audio	5.5	1.8	4.4	6.1	1.3	5.7	14.0	4.0	9.7
Audio Flamingo 3	8.8	2.8	7.6	32.9	21.8	22.6	35.0	24.0	25.0
TimeAudio	7.8	1.8	10.2	39.5	24.9	28.6	21.0	7.0	16.0
TimeAudio + RelTwin	24.25	13.25	18.36	45.63	30.92	34.08	44.00	25.00	30.48
Δ vs. TimeAudio	+16.45	+11.45	+8.16	+6.13	+6.02	+5.48	+23.00	+18.00	+14.48
SpotSound-A	76.5	59.5	57.9	93.4	91.0	85.6	86.0	74.0	69.8
SpotSound-A + RelTwin	78.00	62.00	59.43	93.61	91.01	86.68	89.00	79.00	74.03
Δ vs. SpotSound-A	+1.50	+2.50	+1.53	+0.21	+0.01	+1.08	+3.00	+5.00	+4.23

$\mathcal{L}_{\text{set}} = -\frac{1}{2} \sum_{i,j} t_i(j) \log p_i(j)$ over six answers, replacing $\mathcal{L}_{\text{cand}}$ in Eq. 3. Exchange II swaps targets and their expansions, fixing the span and union.

Replay candidates perturb boundaries, add/drop events, merge spans, split the longest interval, or cover the recording. Stage 2 uses $\mathcal{R}_2 = \mathcal{R}_1 + \lambda \mathcal{L}_{\text{set},r} + \kappa \text{KL}(p_0 \| p_r)$, with frozen base-model distribution p_0 ; $\mathcal{L}_{\text{set},r}$ is the same listwise loss for one replay query. Inference directly generates timestamps.

3.3. Joint localization and relation selection

For interval unions, $\text{IoU}_{\text{set}}(S, G) = |S \cap G| / |S \cup G|$ (zero for an empty union). Given predictions $\widehat{W}_i$, define $u_{ij} = \text{IoU}_{\text{set}}(\widehat{W}_i, W_j)$. PairAcc requires $\min(u_{++}, u_{--}) \geq 0.5$. Swap error occurs if $u_{++} \leq u_{+-}$ or $u_{--} \leq u_{-+}$, including ties. JointPairAcc requires PairAcc success without Swap error. We report pair-level percentages; same-answer counts use exact equality of parsed interval lists.

Predicting the union of two disjoint, equal-length targets for both queries passes PairAcc (IoU 0.5) but fails JointPairAcc. Inspired by paired compositional evaluation [15], JointPairAcc distinguishes query-specific selection from coverage.

4. EXPERIMENTAL SETUP

Data and evaluation protocol. Twin construction uses 40 ESC-50 classes and 256 recordings: 1,024 relation queries (512 pairs) and 512 category queries. Ten held-out classes provide 80 recordings, 320 queries, and 160 inverse pairs

for relation development, with disjoint classes and original Freesound IDs at adaptation. Evaluation covers 400 SpotSound-Bench queries over 387 audios, 6,649 Clotho-Moment queries, and UnAV-100’s 100 queries over 77 audios [2]. We report set-mIoU and R1 at IoU 0.3/0.5, clipping intervals to duration. Relation-development and public scores informed development.

RelTwin training. From official TimeAudio, seed 0 runs 384 Stage 1 and 128 SetPO updates. Its 31.76M trainable parameters use rates 2×10^{-6} (LoRA [16]) and 5×10^{-6} (temporal/audio connectors). Event/relation twins mix 3:1; replay mixes Clotho training, synthetic scale/cardinality, TEMPO training, and AudioGrounding training at 4:3:2:1. We use $\lambda = \gamma = 1$, $\beta = 0.75$, $\kappa = 0.2$, $\tau = 1$, $\tau_q = 0.15$, and 15% interval expansion (minimum 0.05 s).

Matched mechanism experiment. Table 2 isolates $\mathcal{L}_{\text{cand}}$ on SpotSound-A: Eq. 3 with $\gamma = 0$, $\beta = 0.5$, and $\lambda = 0$ (paired SFT) or 1 (candidate supervision), evaluated after Stage 1. Both use identical relation/rehearsal data and 256 AdamW [17] updates at 5×10^{-6} , zero weight decay and gradient clipping 1. LoRA has rank 8, $\alpha = 16$, dropout 0.1, and 20.19M parameters. Each update accumulates one pair and one rehearsal query, sharing initialization, data order, code, and environment. Candidate training adds four gradient-free scores and weighted gradient recomputation to SFT’s two positive-answer backpropagations.

Inference. We use greedy one-beam decoding, capped at 128 new tokens. The mechanism experiment uses 16 kHz mono audio and bfloat16; timing tests use its frozen checkpoints (Section 5.3).

Comparison and statistics. Table 1 compares published

Table 2. Mechanism experiment: isolating RelTwin’s candidate supervision on SpotSound-A. (b) Matched data and updates; (c) mean $\pm$ sample SD. Joint: JointPairAcc; same-answer: inverse-query pairs with identical timestamp answers. Scores are percentages except counts.

Method	SpotSound	Relation development (160 inverse-query pairs)			
	mIoU $\uparrow$	mIoU $\uparrow$	Joint $\uparrow$	Swap error $\downarrow$	Same-answer pairs $\downarrow$
<i>(a) Backbone reference: shared evaluation protocol</i>					
SpotSound-A	58.42	37.64	0.63	65.00	15/160
<i>(b) Matched training comparison</i>					
Paired SFT	58.42	74.58	53.13	40.00	37/160
Candidate supervision	59.43	88.55	84.38	7.50	1/160
Gain over paired SFT	+1.02	+13.97	+31.25	−32.50	−36 pairs
<i>(c) Candidate supervision: three training seeds</i>					
Candidate supervision	59.16 $\pm$ 0.26	87.05 $\pm$ 1.43	81.46 $\pm$ 2.53	8.13 $\pm$ 0.63	(1, 0, 0)/160

baselines; Table 2 tests candidate supervision under shared inference. Mechanism point results use seed 0, selected by SpotSound mIoU from five runs, and matched SFT; Table 2(c) reports seeds 0/1/2 as mean $\pm$ sample SD.

We use 20,000 audio-group bootstrap resamples for 95% CIs; relation tests resample 39 connected source-sharing groups (largest: 18 audios). CIs condition on fixed checkpoints and selection.

5. RESULTS AND ANALYSIS

5.1. Grounding performance across benchmarks

On SpotSound-Bench, Clotho-Moment, and UnAV-100, SpotSound-A + RelTwin reaches 59.43, 86.68, and 74.03 mIoU (Table 1), exceeding its published baseline by 1.53, 1.08, and 4.23 points. TimeAudio + RelTwin reaches 18.36, 34.08, and 30.48, with gains of 8.16, 5.48, and 14.48 points. Both backbones also improve R1@0.5 across all three benchmarks.

On 400 SpotSound queries, candidate supervision gains 1.02 mIoU over matched paired SFT; the audio-group CI is [0.23, 1.81], with 92 wins, 239 ties, and 69 losses. Its three-seed mean is 59.16 (Table 2c). We next examine inverse-query responses.

5.2. Timestamp contrasts recover query-specific selection

Paired SFT returns identical answers for 37/160 inverse-query pairs across 22 recordings; candidate supervision reduces this to one (Table 2). Of those 37 pairs, 27 become jointly correct: the two predictions recover their respective requested relations. Figure 1 illustrates this query-specific recovery, with each inverse query selecting its corresponding event window.

JointPairAcc increases by 31.25 points (source-group CI [22.35, 40.50]) and Swap error decreases by 32.50. Across all pairs, 51 become jointly correct and one regresses. Relation mIoU increases by 13.97 points (CI [9.84, 17.51]), connecting selection with overlap.

Across three seeds, mean JointPairAcc is 81.46; same-answer counts are 1, 0, and 0.

5.3. Binding gains persist after timing intervention

To test layout-specific silence, we recompose 80 development audios with 1.5 s lead-in, 0.25 s within-window gaps, 3 s between windows, and 2 s tail, preserving waveforms, order, and queries. The frozen matched models evaluate one template (160 queries). Timing changes for 40 BA-first recordings; 40 AB-first recordings remain waveform-identical.

On changed recordings, candidate supervision reaches 87.50 JointPairAcc against SFT’s 67.50, a 20-point gain with CI [2.94, 33.33]; originally they scored 85.00 and 42.50. This advantage supports query–window binding across silence layouts. Across all 160 queries, SFT and candidate supervision score 81.34 and 90.65 mIoU.

5.4. Why locally valid answers matter

Shared excerpts hold event identity fixed while inverse queries exchange the correct window. Detecting both categories supports both candidates; selecting the requested order requires query-conditioned comparison. Reduced same-answer collapse and improved JointPairAcc capture this distinction in the controlled binding test. RelTwin extends answer competition through exchange consistency and set-level preferences, retaining direct generation at inference.

6. CONCLUSION

RelTwin combines locally valid timestamp contrasts, exchange consistency, and set-level preferences to learn which occurrence answers a query. Both adapted backbones exceed their published mIoU on three benchmarks. Matched candidate-supervision experiments and timing intervention demonstrate improved query–window binding, while inference preserves direct timestamp generation.

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Compliance with Ethical Standards. We reuse public ESC-50 [14], SpotSound [8], Clotho-Moment, UnAV [2], TEMPO, and AudioGrounding data, including programmatically constructed sequences. No new recordings, participant recruitment, human or animal experiments, or listening studies were undertaken.

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